特集 創発システム
－人工システムの新たなパラダイムをめざして－

解 説

Toward Adaptive Intelligent Systems

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Abstract

The aim of this paper is to provide a perspective for evolutionary computing in the context of research into adaptive intelligent systems. The author proposes a problem class hierarchy, by which problems can be classified into four classes. Traditional AI techniques provide a powerful basis for several classes of problems that can be solved by linear methods or linear approximation. However, solutions to highly nonlinear problems and nonequilibrium environment problems have not yet been established. The author argues that evolutionary computing and emergent computing provide effective problem solving methods for broad range of problems in all classes.

1. Introduction

The legendary strategist Sun Tzu wrote: Provided you know the enemy and know yourself, you need not fear the result of a hundred battles. Despite recent excitement regarding evolutionary computing and artificial life, the impact they are likely to have is still ambiguous. This is due to a lack of analysis of the class of problems we are facing and the potential of the technologies at our disposal. The aim of this paper is to provide a general perspective on the impact of evolutionary computing and artificial life by providing a brief analysis of the nature of the problems to be solved, and clarifying how best to go about solving these problems. Such an analysis would help us deliniate future research directions to be emphasized.

The author argues that a series of recent proposals in evolutionary computing, artificial life, and artificial intelligence offer a new and promising approach to building adaptive intelligent systems. Although the theories and models proposed to date are still premature, these are certainly precursors to a new avenue of research into building intelligent systems.

2. Problem Class Hierarchy

First, we need to know our enemy-the set of problems we have to solve. The problems can be classified into four major categories:

Linear decomposition problems (Class-I): The problem can be decomposed into a set of linear problem. Airline scheduling, and some engineering design problems fall into this class. Expert systems and traditional AI are capable of solving this class of problem effectively.

Linear approximation problems (Class-II): Such problems are essentially nonlinear, but can be solved by linear approximation in practice, allowing for certain levels of error. Limited-domain natural language processing, speech recognition, vision, and many engineering problems fall into this class.

Non-linear problems (Class-III): Such problems are highly non-linear. Thus, they cannot be solved by linear approximation. Many difficult real-world problems, such as full-scale natural language processing, speech recognition, auditory scene understanding, and computer vision come under this class.

Non-equilibrium environment problems (Class-IV): Here, the environment in which the system operates changes dynamically, to the extent that the system's original design must be reformulated. Evolution has proved an effective means of coping with this problem. Deep space exploration and other projects which must cope with unknown and dynamically changing environments are faced with this type of problem.

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To date, artificial intelligence has been applied to problems of class I, II, and III. We already have solutions to classes I and II. Finding a solution to class III is the current focus of attention in the AI community. Work on finding a solution to the class-IV problems has just begun.

3. Linear Problems

Many linear decomposition problems and linear approximation problems can be solved by applying traditional AI techniques and algorithmic methods. Traditional AI techniques, as represented by expert systems, can be characterized by;

- Formal representation
- Rule-driven reasoning
- Strong methods

The basic assumptions of this approach are; experts knew necessary and sufficient knowledge for the task, and this expert knowledge can be expressed in symbolic form. It also assumes that the knowledge acquired is complete, correct, and consistent. Provided these assumptions hold, traditional AI techniques are a powerful means of problem solving. Class-I problem include features that satisfy these assumptions. The salient features of class-I problems are:

Discreteness: Objects and features in the domain have a high degree of discreteness, allowing the domain to be mapped into a symbolic representations.

Explicitness: Rules governing the domain exist in an explicit form.

Completeness: A complete set of rules can be obtained.

For example, SHRDLU was one of the early successes of AI. This system can understand natural language in a block world[30]. Although the manipulation of a real block world is a non-trivial task, Winograd simplified it into an abstract symbol manipulation task. As demonstrated by SHRDLU and other early AI research, traditional AI models are suitable for application to linear problems (Class-I) and linear approximation problems (Class-II).

A number of industrial successes have been reported for the application of AI technologies to real world problems of classes I and II. The XCON project (originally called R 1) is a good example[21]. XCON's task was to apply rules to find a computer configuration to satisfy a symbolically expressed specification. In XCON's domain, all knowledge can be written in a symbolic manner. All components were symbolically labelled. This is a typical class-I problem.

The flight crew dispatch problem is another example of a class-I problem. The objects in this domain are pilot names, equipment, routes, flight numbers, etc. These already constitute a set of discrete symbols, enabling all concepts and objects in the domain to be mapped into symbolic representations. At the same time, airline companies have explicit written rules on how to dispatch flight crews. Such rules were applied even when crew dispatch was done manually. Obviously, these rules are precise and have already been proven to be complete and consistent. There are several real world domains that share the same features. Computer system configuration, logistics planning, and the assembly planning of prefabrication houses are typical examples. In addition, there are many class-II problems that can be effectively solved by applying a traditional AI approach and allowing certain permissible errors.

However, even for the class-I and class-II problems, there are problems which demand immense computing power (often NP-complete) making the straightforward application of traditional methods impractical. The travelling salesman problem (TSP), the bin packing problem, and the network partitioning problem are typical benchmark problems of this kind. These problems share several features with the linear problem, but are not suitable for traditional approaches due to the combinatorial explosion of the search space.

The main advantages of applying the evolutionary computing approach to this type of problems are (1) its ability to provide near-optimal solutions within practical computing costs, and (2) it enables problem solving even when the domain characteristics are not well understood. Although the evolutionary computing approach does not guarantee that an optimal solution can be obtained, a

near-optimal solution suffices in most practical cases. There are several reports of an efficient solution being obtained using genetic algorithms. However, it should be noted that there are many studies that revealed that evolutionary approach is far inefficient than optimized algorithmic approaches when tasks are well investigated.

4. Non-linear Problems

Class-III problems involve a highly nonlinear solution space. **Fig. 1** (left) illustrates the problem of how to describe a nonlinear region symbolically. Typically, such an area would be described as (and A 2 (or B 2 B 3)). However, this symbolic representation corresponds to the gray area in Fig. 1 (left), making certain errors inevitable. Subdividing symbols into smaller pieces would reduce errors, but excessive subdivision could results in the semantics of symbols being lost. Even an attempt to describe a simple two-dimensional non-linear area is difficult when taking a symbolic approach. The real world entails a far more complex high-dimensional and time-varying solution space.

To find a solution to Class-III problems, a re-examination of the theory of symbol and sensory perception is important. Traditional AI models assume (1) global mapping of knowledge and sensory data, and (2) passive perception. New approaches would be based on local mapping and active perception.

Local Mapping The main reason why the traditional symbolic approach fails to accurately cover nonlinear regions is because it relies on the global mapping of data. The opposite approach is local mapping. Local mapping attempts to obtain information, in these parts of the region relevant to the processing. Classical version of this approach is known as a picewise linear mapping. The multiple probing of local regions would provide a highly accurate description of the region (Fig. 1 (right)). Memory-based reasoning and memory-based translation approaches employ this strategy, thereby attaining highly accurate reasoning, especially when compared with traditional methods[26),27)].

Active Perception One other characteristic of the traditional AI approach is its control structure. **Fig. 2** illustrates three different information processing models in AI. The SMPA model represents the traditional AI approach, where the processing follows a Sense-Model-Plan-Act sequence. An alternative approach is the subsumption architecture, where a bottom-up approach is applied to the problem[6),7)]. In either cases, however, agents percieve data and react against the sensory inputs. In a sense, the central system is a slave to the peripherals.

Active perception makes radical departure from the traditional information processing models. In active perception, the agent selects which data to obtain and actively percieves its environment (Fig. 2 (right)). In other words, the system must be able to control the focus of its attention. Several recent research projects have already taken this approach. Examples include deictic representation[2)], active vision[4)] and visual routing[8)]. the ideas of which this research is based are "the internal representation of the image is minimal with respect to the goals"[5)], and "A visual routine is task-specific and performed only when it is useful[8)]. Active perception is consistent with local mapping because active perception assumes that the agent is capable of selecting the data to percieve.

For the system to actively select and percieve its environment, the system must have goals. Other-

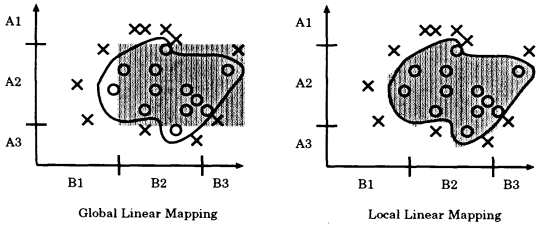


Fig. 1 Global Mapping vs. Local Mapping

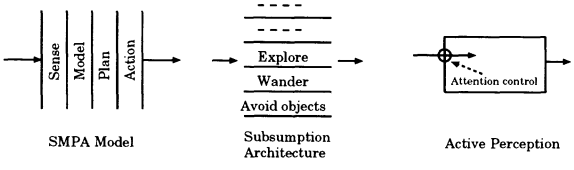


Fig. 2 Information processing in various models

wise, the system can not determine what it must percieve. Obviously, some research on situated agents focuses on how the system can have goals and how to handle these goals. If the system's goals, the importance of each item of data, and the key features of the data set can be predefined and understood, building the active perception system need not be difficult. Unfortunately, this assumption does not hold true in many cases. Perhaps the solution is to apply evolution to acquire the evaluation function to provide a focus of attention. Thus, in some cases, the solution to a class-III problem can be derived from the stationary solution to a Class-IV problem. This will become clear in the next section.

5. Non-equilibrium Environment Problems

The next challenge is to tackle the dynamically changing environments that are typically observed in nonequilibrium open systems. This is a Class-IV problem. In this class, the system operates in an environment where the characteristics of the environment change drastically, such that the system's optimization function itself may have to be revised.

For example, suppose that a certain type of insect provides an excellent source of nutrition to a given spiece. An individual that can quickly learn to capture and eat the insect may stand a better chance of surviving and producing offspring. Within this enviroment, the optimal learning strategy is to maximize the capture of that type of insect. However, if the insect has evolved such that it is poisonous to its predators, that learning strategy is no longer optimal. The system must revise its learning goal such that it can avoid those poisonous insects and instead capture an edible variety. The evolution of species has been a history of adapting to a dynamically changing environment.

Theoretically, systems with such dynamics can be classified into fixed point attractors, limit cycles, or chaos. The real world, however, is most likely to exhibit chaos. Thus implies that the prediction of a system's learning goal is virtually impossible. This type of environmental change is not limited to biological systems. A broad range of social and economic systems are now recognized as exhibiting similar behavior. Thus, I argue that certain levels of adaptation capability will be required by a wide range of next-generation systems.

Evolutionary computing provides a powerful means of continuously optimizing a system's learning goal through adaptation. Evolutionary Reinforcement Learning (ERL[1]) was an early attempt at applying evolution to the acquisition of an evaluation function in reinforcement learning. However, the ERL model does not incorporate learning modulation when some events are more important to survival than other events. This aspect was modelled in[7] as hormonal modulation of learning. In its simplest form, the evaluation function evolves to adapt itself to the environment and provide appropriate reinforcement signals to modulate learning.

6. Emergence and Evolution of Intelligence

This paper has concerned itself with the characteristics of problems in each of four classes, and how current technologies can be used to attack these problems. However, actually building an adaptive and intelligent system is a non-trivial task. The author predicts that developing an intelligent system and making it adaptive would merely repeat the mistake of traditional AI, which started with a toy system that researchers subsequently attempted to scale-up. In much the same way as nonlinear problems that can be classified into Class-II, some nonequilibrium environment problems can be redefined to approximate to equilibrium environment problems that can be classified into Class-I, -II, or -III problems. This can be accomplished by imposing certain restrictions on the system operating conditions such as the duration of the operation. However, without such restrictions, the system designer is unable to predict the operating environment because of the chaotic nature of the real world.

The question becomes one of how can we build an intelligent system that is highly adaptive, such that it can cope with a non-equilibrium environment. One feasible path is to make a system adaptive and them increase its intelligence (**Fig. 3**). Obviously, following this path is not a trivial task, either. However, any attempt to increase the adapt-

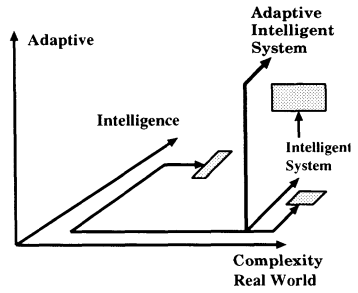


Fig. 3 A Path to Adaptive Intelligent Systems

ability of an existing system would be far more difficult.

It is hard to say just how complex and adaptive a system can be built through evolution. For example, any intelligent system should be able to use language, which is a highly complex symbol manipulation system. The issue is one of how circuits acquired by evolution climb up the ladder of intelligent behavior. For example, one way of classifying the evolution of language is:

No Symbol (Level-0) : No effective language exists.

Single Token (Level-1) : A very primitive form of language, or communication via chemical substances such as feromones.

Token Combination (Level-2) : The power of expression increases by combining a number of primitive signs. Apes appear to use this level of language.

Embedding (Level-3) : The power of expression increases drastically as this class of language allows complex structure operations.

Recursion (Level-4) : Maximum expressive power can be obtained by recursion. Human languages are of this level.

There is a need for serious investigation into how evolution forced the development of the complex neural circuits that enable us to use complex language[12].

One other question is whether a new theory of symbol grounding can be formulated. One possible consequence of employing active perception is the view of "symbol as emergent observable" In the traditional model, the symbol exists regardless of how the cognitive agent percieves the world. However, this is not necessary in the case of active perception. Symbols can be treated as emergent observables that emerge only when the observer observes the world. The author previously propose the quantum symbol hypothesis as a working hypothesis for describing this aspect of the symbol grounding problem. Serious investigation has been performed on the chaotic nature of neural activities[24),28)], cortical circuits[25)], self-organization[3),15)], and symbol dynamics[11)]. The results obtained must be linked with biological findings[9),15)].

Overall, in the short run, evolutionary computing provides a wide range of effective problem solving methods for all classes of problems. Looking at a longer time frame, a series of scientific discoveries are awaited in the interdisciplinary areas of computer science, biology, and the physics of complex systems.

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会 員 の 皆 様 へ

毎号の巻末にある「さいすらん」をご愛読いただいていることと思います．これは会員の皆様のご意見を掲載するコラム欄ですが，投稿にはこれまでは，原稿の学会への郵送とファクシミリによる2つの方法がありました．今月から，これに加えまして電子メールによる投稿が可能になりました．

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です．原稿の通常のサイズは半ページ(960文字を目安にしてください)ですが，これよりも短いものでも結構ですのでお気軽に投稿してください．

これからもよい「さいすらん」にするよう工夫していくつもりです．ご意見・ご要望などがございましたらお知らせください．